

MAUSAM 2.0: Next-Gen Personalized Weather System

Adaptive, Multi-Persona Homepage & Proactive Meteorological Intelligence Platform

PROBLEM STATEMENT ID

26076

MINISTRY / DEPARTMENT

Ministry of Earth Sciences (MoES)
India Meteorological Department (IMD)

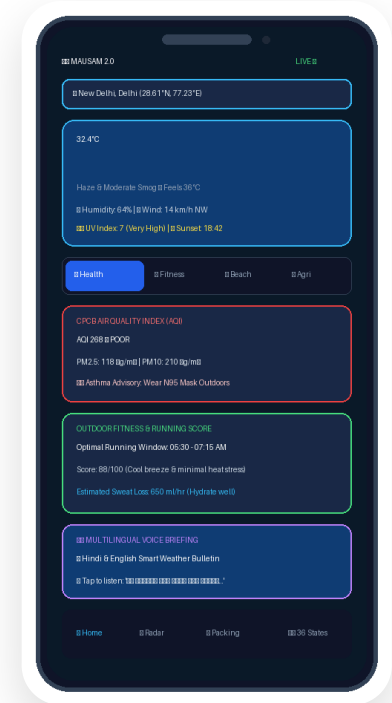
THEME & PS CATEGORY

Theme: Smart Automation | Category: Software

PROBLEM STATEMENT TITLE

Development of personalized homepage for 'Mausam' mobile application

Team ID: [Your Team ID] | Team Name: [Your Registered Team Name] | Team Leader: [Leader Name]



Problem Statement & Proposed Solution

Bridging the gap between raw meteorological data and actionable citizen lifestyle decisions



Current Bottlenecks in Weather Apps

- **Generic Weather:** Farmers, athletes, travelers receive the exact same static temperature card.
- **Information Overload:** Users receive raw atmospheric values (68% RH, 1008 hPa) without activity context.
- **No Actionable Insights:** Data does not translate into advice: best workout time, school commute, or spraying.
- **Static Homepage:** The homepage remains rigid instead of adapting dynamically to daily routines.



Proposed Solution: Mausam 2.0

- Adaptive, persona-driven platform** converting meteorological data into contextual, actionable guidance.
- **8 Tailored Personas:** Health, Fitness, Beach, Travel, Family, Agri, Commute, Events.
 - **Automated Models:** WBGT heat stress, running score, tide harmonics, soil moisture %, packing checklists.
 - **Pan-India Reach:** Directory for all 28 States & 8 UTs + GPS reverse geocoding.
 - **Multilingual Voice:** Natural Text-to-Speech (TTS) bulletins in English & Hindi.



8 Persona Intelligence Modules + Smart Auto-Blend

Contextual dashboards tailored to specific user activities, vulnerability factors, and routines

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Health-Conscious

CPCB AQI (PM2.5/10), Pollen risk, UV sunburn timer, asthma alerts.



Outdoor Fitness

Best Running Hours (0-100), WBGT heat index, sweat hydration rate.



Beach & Surfers

Sea condition status, 24h Tide harmonic curve, swell wave height (m).



Travelers

Multi-city comparator, airport fog delay risks, AI packing checklist.



Parents & Family

School commute windows (7-9 AM & 2-4 PM), rain radar, playground.



Agriculture / GKMS

Topsoil (0-7cm) moisture %, 7-day crop rain balance, spraying window.



Daily Commuters

Road visibility distance (meters), urban waterlogging risk, traction.



Event Planners

14-day rain probability, Thermal Comfort Index, wind load gust ratings.



SMART AUTO-BLEND (AI + ROUTINE ENGINE): Dynamically transitions homepage: Morning Fitness (06-08:30) → Commute (08:30-10:30) → School Transit (14-16:00) → Evening Leisure (18-21:30).

Technical Architecture & System Methodology

Modular, scalable, zero-dependency full-stack meteorological pipeline implemented in code

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1. Data Ingestion & API Gateway

- **server.py (Python 3):** Standard library async HTTP proxy server for weather, AQI & geocoding.
- **weatherService.js:** Aggregates synoptic forecasts, CPCB AQI, INCOIS marine waves & soil moisture.
- **Pan-India Geocoding:** 36 States/UTs directory + OpenStreetMap reverse geocoding for GPS.
- **5-Min Cache:** Memory caching eliminating redundant external API queries.



2. Persona Intelligence Engine

- **personaEngine.js:** Core dispatch module with calculations & widgets for all 8 personas.
- **WBGT & Thermal Models:** Computes heat stress and sweat-rate hydration loss (ml/hr).
- **Tidal Harmonics Model:** Generates 24h sinusoidal tidal curves and peak high/low tide times.
- **packingAdvisor.js:** AI travel packing matrix analyzing temperature deltas & rain for gear lists.



3. Client Experience & Voice TTS

- **Universal PWA Client:** Responsive Tailwind CSS UI with 1-click Mobile Shell simulator.
- **charts.js Visualization:** Canvases for dual-axis time series, AQI bars, and crop moisture curves.
- **voiceBriefing.js:** Web Speech API engine synthesizing voice briefings in English & Hindi.
- **app.js Orchestrator:** Master state management, local storage caching, and GPS handler.

Innovation, Feasibility & Expected Impact

Delivering high-value public safety, agricultural efficiency, and citizen engagement

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Key Innovations

- **Smart Auto-Blend:** The homepage dynamically adapts across daily routines.
- **Persona Intelligence:** Raw weather data transformed into specialized insights.
- **AI Smart Packing:** Destination forecasts generate tailored packing checklists.
- **Actionable Guidance:** Concrete recommendations (spray window, N95 mask).
- **Multilingual Voice:** Built-in TTS in English & Hindi improves accessibility.



Technical Feasibility

- **Zero Dependency Backend:** Python 3 standard library; runs cleanly on any basic VM.
- **Modular Architecture:** Clean separation of data, persona, chart, and voice layers.
- **High-Speed Caching:** 5-minute memory caching minimizes outbound API queries.
- **Offline-First Resilient:** LocalStorage caching maintains full offline functionality.
- **Seamless Deployment:** Lightweight design compatible with PWA, Android & Webview.



Expected Impact

- **Public Health:** Correlates humidity with PM2.5 to provide asthma & heatstroke warnings.
- **Agriculture:** Helps farmers make weather-informed irrigation & spraying decisions.
- **Travel & Transit Safety:** Reduces travel packing uncertainty & gives visibility alerts.
- **Citizen Experience:** Transforms complex meteorology into clear recommendations.
- **Accessibility:** Audio briefings ensure elderly & rural users receive weather updates.

Roadmap, Scalability & Future Scope

Phased deployment strategy for national rollout across 1.4 billion citizens (Proposed Future Scope)

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Phase 1: Platform Expansion

Timeline: 0 - 3 Months

- Deploy Mausam 2.0 PWA across Android, iOS, and Web platforms.
- Expand weather data integrations with IMD NOWCAST push notifications.
- Enhance recommendation quality and user preference customization.
- Expand multilingual voice briefing to 12 regional Indian languages.
- Strengthen offline mobile caching and PWA installability.



Phase 2: Smart Alerts & Integration

Timeline: 3 - 6 Months

- Integrate with NDMA Common Alerting Protocol (CAP) for disaster alerts.
- Enable crowdsourced hyper-local weather & waterlogging reporting.
- Integrate with IRCTC & Flight booking portals for travel packing alerts.
- Provide district-level push notifications for extreme weather events.



Phase 3: Hyper-Local Intelligence

Timeline: 6 - 12 Months

- Implement on-device Edge ML for offline micro-climate forecasting at 1km.
- Integrate with Gramin Krishi Mausam Sewa (GKMS) village IoT sensors.
- Provide Smart-Home and Wearable integration (WearOS / Apple Watch).
- Develop AI-driven personalized micro-climate forecasting models.